

Google Company Preparation Syllabus & 6-Month Plan

Section 1: Google Preparation Syllabus

1. Data Structures & Algorithms (DSA)

- Arrays, Strings, Linked Lists, Trees, Graphs, Heaps
- Dynamic Programming, Greedy, Backtracking
- Sorting, Searching, Bit Manipulation

2. Computer Science Core Subjects

- Operating System: Threads, Processes, Scheduling, Deadlock
- DBMS: Normalization, Transactions, Indexing, SQL
- Networks: TCP/IP, OSI, Routing, HTTP, DNS
- OOP: Inheritance, Polymorphism, Design Patterns

3. System Design

- High-Level Design: Load Balancers, Databases, Caches
- Low-Level Design: Class Design, APIs, Object Modeling

4. Coding Platforms

- LeetCode (Top 300), HackerRank, Codeforces

5. Machine Learning (For ML Roles)

- Supervised/Unsupervised Learning, Neural Networks
- Sklearn, TensorFlow, Feature Engineering, Deployment

6. Behavioral Interviews

- Googleyness, Team Fit, STAR Method

7. Resume & Projects

- Impactful Projects, Problem Statements, Results

Section 2: 6-Month Weekly Plan

Month 1:

- DSA Basics (Arrays, Strings, HashMap, Stack, Queue)
- OS Concepts + SQL Queries

Month 2:

- Linked Lists, Recursion, Sorting Algorithms
- DBMS + OOPs in depth

Month 3:

- Trees, Graphs, Dynamic Programming
- Low-Level Design (Classes, Relationships, APIs)

Month 4:

- System Design (HLD: Load Balancers, Caches, DBs)
- Practice Mock Interviews (Daily 1 Problem Solving)

Month 5:

- ML Algorithms (Linear, Tree, Clustering, NN)
- Model Deployment + Real-world ML Projects

Month 6:

- Final Revisions of All DSA Topics
- Resume Finalization + Mock Interviews
- Behavioral Prep (STAR Stories, Googleyness)